Importing Libraries

In [1]:
```
import pandas as pd
```

Reading our CSV File

In [2]:
```
import requests
url = 'https://raw.githubusercontent.com/codebasics/py/master/ML/9_decis
ion_tree/Exercise/titanic.csv'
res = requests.get(url,allow_redirects = True)
with open('titanic.csv','wb') as file:
    file.write(res.content)
```

In [3]:
```
df = pd.read_csv('titanic.csv')
df.head()
```

Out[3]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Far
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.283
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.100
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500

In [4]:
```
df['Age'].unique()
```

Out[4]:
```
array([22.  , 38.  , 26.  , 35.  ,   nan, 54.  ,  2.  , 27.  , 14.  ,
        4.  , 58.  , 20.  , 39.  , 55.  , 31.  , 34.  , 15.  , 28.  ,
        8.  , 19.  , 40.  , 66.  , 42.  , 21.  , 18.  ,  3.  ,  7.  ,
       49.  , 29.  , 65.  , 28.5 ,  5.  , 11.  , 45.  , 17.  , 32.  ,
       16.  , 25.  ,  0.83, 30.  , 33.  , 23.  , 24.  , 46.  , 59.  ,
       71.  , 37.  , 47.  , 14.5 , 70.5 , 32.5 , 12.  ,  9.  , 36.5 ,
       51.  , 55.5 , 40.5 , 44.  ,  1.  , 61.  , 56.  , 50.  , 36.  ,
       45.5 , 20.5 , 62.  , 41.  , 52.  , 63.  , 23.5 ,  0.92, 43.  ,
       60.  , 10.  , 64.  , 13.  , 48.  ,  0.75, 53.  , 57.  , 80.  ,
       70.  , 24.5 ,  6.  ,  0.67, 30.5 ,  0.42, 34.5 , 74.  ])
```

In [5]:
```
#df.dropna(inplace=True)
df.Age.fillna(df.Age.mean(),inplace=True)
```

In [6]:
```
df['Age'].unique()
```

Out[6]:
```
array([22.        , 38.        , 26.        , 35.        , 29.69911765,
       54.        ,  2.        , 27.        , 14.        ,  4.        ,
       58.        , 20.        , 39.        , 55.        , 31.        ,
       34.        , 15.        , 28.        ,  8.        , 19.        ,
       40.        , 66.        , 42.        , 21.        , 18.        ,
        3.        ,  7.        , 49.        , 29.        , 65.        ,
       28.5       ,  5.        , 11.        , 45.        , 17.        ,
       32.        , 16.        , 25.        ,  0.83      , 30.        ,
       33.        , 23.        , 24.        , 46.        , 59.        ,
       71.        , 37.        , 47.        , 14.5       , 70.5       ,
       32.5       , 12.        ,  9.        , 36.5       , 51.        ,
       55.5       , 40.5       , 44.        ,  1.        , 61.        ,
       56.        , 50.        , 36.        , 45.5       , 20.5       ,
       62.        , 41.        , 52.        , 63.        , 23.5       ,
        0.92      , 43.        , 60.        , 10.        , 64.        ,
       13.        , 48.        ,  0.75      , 53.        , 57.        ,
       80.        , 70.        , 24.5       ,  6.        ,  0.67      ,
       30.5       ,  0.42      , 34.5       , 74.        ])
```

In [7]:
```
inputs = df.drop('Survived',axis=1)
inputs.head()
```

Out[7]:

	PassengerId	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	E
0	1	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

In [8]:
```
target = df['Survived']
target
```

Out[8]:
```
0      0
1      1
2      1
3      1
4      0
      ..
886    0
887    1
888    0
889    1
890    0
Name: Survived, Length: 891, dtype: int64
```

Encodeing 'Sex' feature

In [9]:
```
from sklearn.preprocessing import LabelEncoder
```

In [10]:
```
le_sex = LabelEncoder()
```

In [11]:
```
inputs['sex_n'] = le_sex.fit_transform(inputs['Sex'])
inputs.drop(['Sex','Name','Ticket','Cabin','Embarked','PassengerId'],axi
s=1,inplace=True)
inputs.head()
```

Out[11]:

	Pclass	Age	SibSp	Parch	Fare	sex_n
0	3	22.0	1	0	7.2500	1
1	1	38.0	1	0	71.2833	0
2	3	26.0	0	0	7.9250	0
3	1	35.0	1	0	53.1000	0
4	3	35.0	0	0	8.0500	1

Splitting Data into Training and Testing

In [12]:
```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test = train_test_split(inputs,target,test_size
=0.2)

print("Shape of x_train :", x_train.shape)
print("Shape of y_train :", y_train.shape)
print("Shape of x_test :", x_test.shape)
print("Shape of y_test :", y_test.shape)
```

```
Shape of x_train : (712, 6)
Shape of y_train : (712,)
Shape of x_test : (179, 6)
Shape of y_test : (179,)
```

Train Our Decision Tree Classifier

In [13]:
```
from sklearn.tree import DecisionTreeClassifier
tree = DecisionTreeClassifier()
```

In [14]:
```
tree.fit(x_train,y_train)
```

Out[14]:
```
DecisionTreeClassifier(class_weight=None, criterion='gini', max_depth=No
ne,
            max_features=None, max_leaf_nodes=None,
            min_impurity_decrease=0.0, min_impurity_split=None,
            min_samples_leaf=1, min_samples_split=2,
            min_weight_fraction_leaf=0.0, presort=False, random_state=No
ne,
            splitter='best')
```

Accuracy Of Our Model

In [15]:
```
tree.score(x_test,y_test)
```

Out[15]:
```
0.770949720670391
```